

**Third Semester B.Tech (Mechanical) (CBCS) Degree Examinations**

ME1007-1 – MATERIAL SCIENCE AND ENGINEERING

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. The atomic packing factor in a simple cubic unit cell is _____.
A) 0.74
B) 0.52
C) 0.68
D) 0.66
2. Amorphous solids have _____ structure.
A) Regular
B) Linear
C) Irregular
D) Dendritic
3. Which type of defect are point defects?
A) One dimensional defect
B) Zero dimensional defect
C) Two dimensional defect
D) Three dimensional defect
4. Which type of stress is involved in the motion of screw dislocation?
A) Tensile stress
B) Shear stress
C) Compressive stress
D) Does not involve stress
5. Which of the following cannot be obtained using a phase diagram?
A) Melting temperatures of various phases
B) Temperature range for solidification
C) Equilibrium solid solubility
D) Purity of materials
6. The maximum concentration of solute that can be added is defined as _____.
A) Solution limit
B) Solubility limit
C) Concentration
D) Degrees of freedom
7. How is Gibb's phase rule defined?
A) $C+P+1$
B) $C+P+2$
C) $C-P+2$
D) $C-P$
8. Which of the following characteristics is mainly considered for the solidification of castings?
A) Appearance
B) Crystal structure
C) Moulding capacity
D) Surface finish
9. Alloys containing 2.0-6.7% carbon are considered as _____.
A) Steel
B) Aluminum
C) Cast-iron
D) Brass
10. How much iron does cementite contain?
A) 6.67%
B) 24.23%
C) 73.4%
D) 93.3%
11. What is the maximum solubility of carbon in austenite at 1148°C?
A) 0.09%
B) 0.8%
C) 2.08%
D) 14.35%
12. The first step in constructing a TTT diagram involves _____ the sample.
A) Annealing
B) Normalizing
C) Quenching
D) Austenising
13. Casehardening refers to _____ the material, after which necessary heat treatment is done to it.
A) annealing
B) critical range
C) carburizing
D) hardening

14. In normalizing of hypoeutectoid steel, the component is heated to a temperature above
 A) 535°C B) 610°C
 C) 800°C D) 910°C
15. Which of the following processes will one use on hardened steel to reduce brittleness?
 A) Annealing B) Normalizing
 C) Spheroidizing D) Tempering
16. For steel, which one of the following properties can be enhanced upon annealing?
 A) Hardness B) Toughness
 C) Ductility D) Resilience
17. In which of the following stages do the deformation rate increases and causes failure?
 A) Transient creep stage B) Constant creep stage
 C) Fracture stage D) Steady stage creep stage
18. Which of the following is costliest among the non ferrous materials?
 A) Magnesium B) Aluminum
 C) Titanium D) Copper
19. Which of the following is the lightest among the following?
 A) Magnesium B) Aluminum
 C) Titanium D) Copper
20. Composite can be classified on the basis of:
 A) Matrix type B) Reinforcement constituent
 C) Matrix type & Reinforcement constituent D) None of the mentioned

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define with example (i) Unit cell (ii) Atomic packing factor.	4	L*1	1	1
	b) Give the detailed classification of engineering materials.	4	L1	1	1
	c) What is a solid solution? Mention the types of solid solution.	8	L1	2	1
2.	a) What is a surface defect? Explain types of Surface defects with diagrams.	8	L2	1	1
	b) With neat diagram explain cast metal structure.	4	L1	2	1
	c) Explain with sketch Free energy curve.	4	L1	2	1
3.	a) Differentiate slip and twinning.	4	L1	1	1
	b) Explain melting of metals with graph.	4	L1	1	1
	c) Two metals A and B have their melting points at 900°C and 800°C respectively. The alloy pair forms a eutectic at 600°C of composition 60% B and 40% A. A and B have unlimited mutual liquid solubilities. Their solid solubilities are as follows: 10% B in A at 600°C and 5% B in A at 0°C 8% A in B at 600°C and 4% A in B at 0°C Assume the liquidus, solidus and solvus lines to be straight. No solid state reactions or any intermediate phase changes occur in the series, i) Draw the phase diagram for the series and label all salient temperatures, compositions and regions. ii) Find the room temperature (25°C) structure of any alloy of composition 60% A and 40% B, with respect to the number, type, extent and composition of the phases.	8	L3	2	1
Unit – II					
4.	a) What is T.T.T diagram? Explain with a neat diagram the Martensitic transformation of Austenite.	8	L2	3	1
	b) Explain Martempering and Austempering heat treatment processes with neat diagram.	8	L2	4	1

5.	a)	What is hardenability? Explain Jominy End-Quench Test with neat sketch.	8	L2	3	1
	b)	Write a note on Surface treatments. Explain the Induction hardening process with neat sketch.	8	L2	4	1
6.	a)	Draw a neat sketch of Fe-Fe ₃ C equilibrium diagram and briefly explain.	8	L2	3	1
	b)	Explain Annealing and Normalizing heat treatment processes with sketches.	8	L2	4	1

Unit – III

7.	a)	Draw and explain types of fatigue cycles.	4	L1	5	1
	b)	List linear and non- linear properties.	4	L1	5	1
	c)	Explain the role of matrix and reinforcement in composite materials.	4	L1	5	1
	d)	Write the types and properties of magnesium alloys.	4	L1	5	1
8.	a)	Explain with the sketch R.R. Moore's fatigue test and draw S-N curve.	8	L2	5	1
	b)	Write a note on Nickel and its alloy materials.	4	L1	5	1
	c)	Briefly explain reinforcement materials used in composites.	4	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
